Faculty of Technical Sciences, University of Novi Sad

Technical Dossier 1.0.0

Bolnica 1.0.0

Hospital Management Suite

Kórházkezelő rendszer

Dossier 1.0.0 · software 1.0.0 · frozen 21 June 2021

A record of the 2021 WPF hospital suite as it actually ran, written so a later rewrite has something solid to start from. Defects are listed on purpose.

Compiled 7 September 2026

Git commit 5b69c9c

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Document control

Field	Value
Title	Technical Dossier 1.0.0, Bolnica 1.0.0
Dossier version	1.0.0
Software version	1.0.0 (archive label; login window title is still `MainWindow`)
Development	16 March 2021 to 21 June 2021
Main coding stretch	late March to 10 June 2021 (heaviest clean-code week: 10 June)
Git freeze	`5b69c9c` (21 June 2021)
First commit	`8fe447c`, Initial commit (16 March 2021)
Commits on main	305
Original upstream	github.com/kukuckarastislav/SIMS_IS_Bolnica
Archive remote	github.com/laslo-uri/HospitalManagementSuite
Freeze path	`_legacy-1.0.0/` (BolnicaProjekat, Dijagrami, Revizije)
Compiled	7 September 2026
Authors	Rastislav Kukučka (RA 129/2018), Mihajlo Ostojić (RA 134/2018), Laslo Uri (RA 135/2018), Milica Vučinić (RA 139/2018)
GitHub	github.com/laslo-uri, github.com/mihajlo-ostojic, github.com/kukuckarastislav
Institution	Faculty of Technical Sciences, University of Novi Sad, Computing and Control Engineering

1. Purpose

This is the **1.0.0** technical dossier. It describes Bolnica at commit `5b69c9c`, parked under `_legacy-1.0.0/`. It is meant to be used when the system is rebuilt: what the old desktop suite did, how it was put together, and what should not be copied.

It covers identity, behaviour as coded, architecture, the live 2021 UI (screenshots from a 2026 run), persistence, validation, and the defects.

On 7 September 2026 the archive was restored with NuGet, compiled with VS 2022 Build Tools MSBuild (`.NET Framework 4.7.2` reference assemblies via NuGet when the machine targeting pack was missing), launched from `bin\Debug`, and exercised with the seed accounts. Screenshots below are that run. The PowerDesigner drawings under `_legacy-1.0.0/Dijagrami/` and the peer notes under `_legacy-1.0.0/Revizije/` stay as course artifacts.

2. Identification

Item	Detail
Display name	Bolnica (assembly `Bolnica`; login title `MainWindow`)

Item	Detail
English name	Hospital Management Suite
Serbian name	Bolnica / Zdravo Bolnica (secretary chrome)
Hungarian name	Kórházkezelő rendszer
Solution	`_legacy-1.0.0/BolnicaProjekat/Bolnica/Bolnica.sln`
Projects	`Bolnica` (WinExe), `Jarloo.Calendar` (calendar control)
Entry	`Bolnica.App` → `MainWindow`
Kind	Multi-role single-process desktop GUI
UI toolkit	WPF (XAML), MaterialDesignThemes, Extended.Wpf.Toolkit, LiveCharts.Wpf
Target	.NET Framework **4.7.2** (calendar project: 4.0)
PDF / charts	iText7 7.1.15, LiveCharts 0.9.7
JSON	`System.Text.Json` 5.0.2
Persistence	Nineteen JSON files under `_legacy-1.0.0/BolnicaProjekat/Bolnica/podaci/`
UI language	Serbian copy throughout

Author	Index	Course role	GitHub
Rastislav Kukučka	RA 129/2018	Upravnik	github.com/kukuckarastislav
Mihajlo Ostojić	RA 134/2018	Sekretar	github.com/mihajlo-ostojic
Laslo Uri	RA 135/2018	Lekar	github.com/laslo-uri
Milica Vučinić	RA 139/2018	Pacijent	—

On `main`, git reports `kukuckarastislav` (84), `mihajlo-ostojic` (78), `laslo-uri-ftn` (77), `Milica Vucinic` (50), `milicauns` (16). Messages are mostly Serbian. The tip message is `test - git nalog`.

3. Historical context

This is the usual FTN SIMS + HCI hospital exercise: four actors, appointments, prescriptions, rooms and equipment, patient self-service, JSON instead of SQL. Course UML was drawn in PowerDesigner; the product is one WPF executable with four role homes after login.

Date	Event
16 Mar 2021	Repository created; roles listed; use-case diagrams
21–28 Mar 2021	Class diagrams per role; use-case fixes
Apr–May 2021	Role UIs, repositories, seed JSON, HCI work

Date	Event
early Jun 2021	Activity / sequence diagrams; peer reviews (KT4)
10 Jun 2021	Heavy clean-code / SOLID / factory / DTO refactors across roles
21 Jun 2021	Tip commit `test - git nalog`. No later functional work in this archive.
7 Sep 2026	Freeze parked under `_legacy-1.0.0/`; dossier compiled and UI captured

Spring 2021 project. The last functional week is almost all refactoring, not new features.

4. Scope

The program is a **local multi-role desktop app**: one process, one login window, four Serbian shells that share in-memory repositories and the same `podaci/*.json` files. Controllers call services; services call singleton repositories; repositories rewrite JSON.

It is not four separate installers, not a networked HIS, not SQL, not HTTPS, and passwords are plaintext. Concurrent multi-user access to the same folder is not implemented (in-process singletons only). Marketing copy that speaks of "four apps" means four login personas on **one** `Bolnica.exe`.

5. The 2021 user interface

Screens captured from this archive on 7 September 2026 after a successful Debug build. They show the freeze. They are not a design target for a new version.

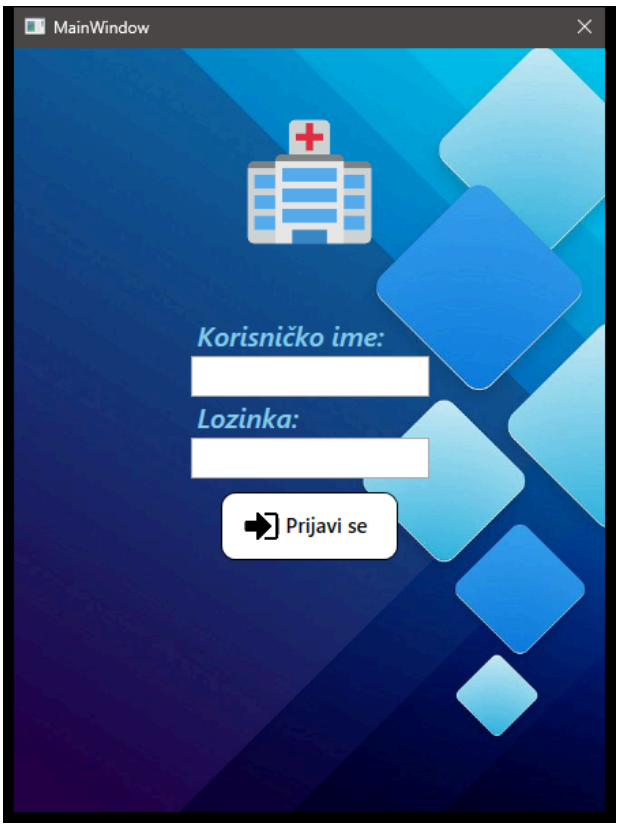


Figure 1. Login (`MainWindow`). Cyan gradient background, hospital icon, labels Korisničko ime / Lozinka, button Prijavi se. Window title remains MainWindow.



Figure 2. LekarHome — Glavni meni for Laslo Uri (nema specijalizaciju). Maroon "mobile" shell: Radni kalendar, Lista pacijenata, Lista lekova, Slanje povratnih informacija.



Figure 3. Radni kalendar — Današnji / Nedeljni / Mesečni tabs. Današnji list empty on this seed clock (no appointments for “today” in 2026).



Figure 4. Lekovi area after menu navigation (same narrow shell).



Figure 5. SekretarHome — Zdravo Bolnica. Dark sidebar (registracija, liste, Zakaži termin, Obavestenja, Zauzetost prostorija). First-run wizard overlay with Serbian typos (aplikacijum, dogme, ocenivanj).



Figure 6. Secretary shell after attempting schedule navigation (sidebar still visible).

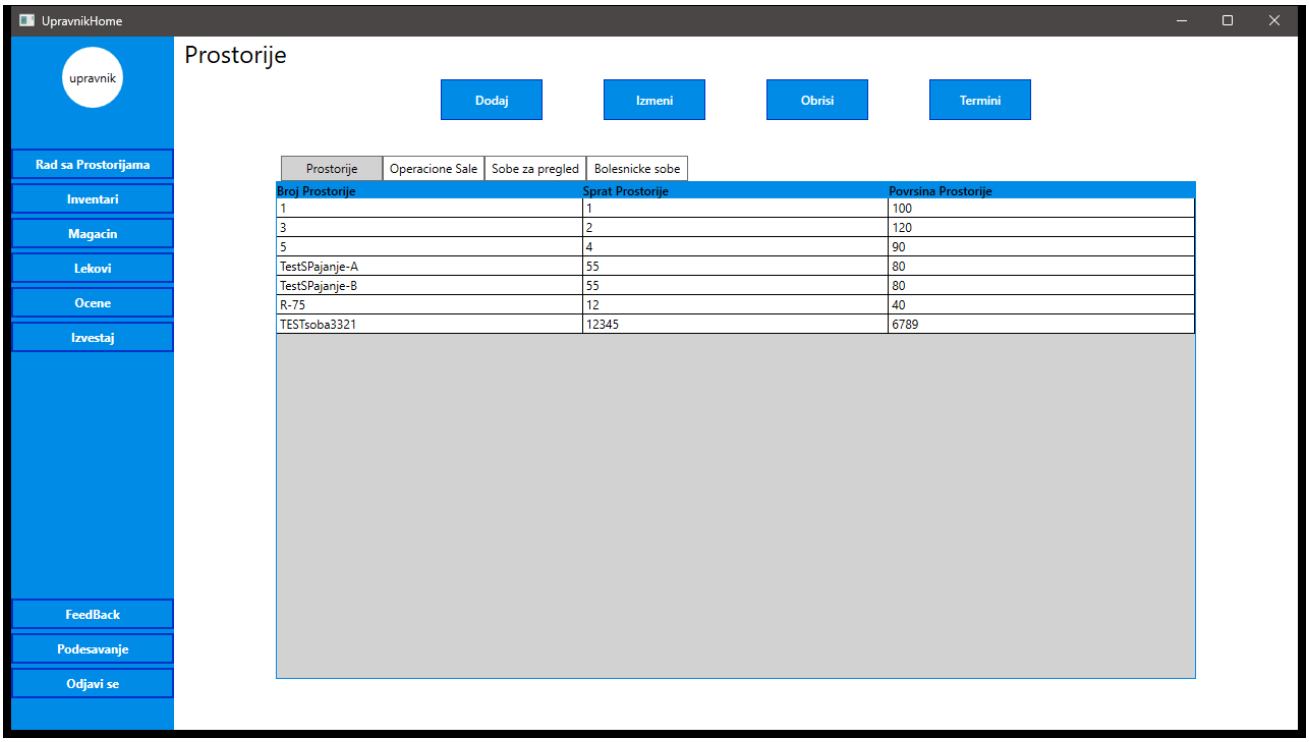


Figure 7. UpravnikHome — Prostorije CRUD. Tabs for room types; seed includes junk rows (TESTsoba3321 floor 12345, TestSPajanje-A/B).

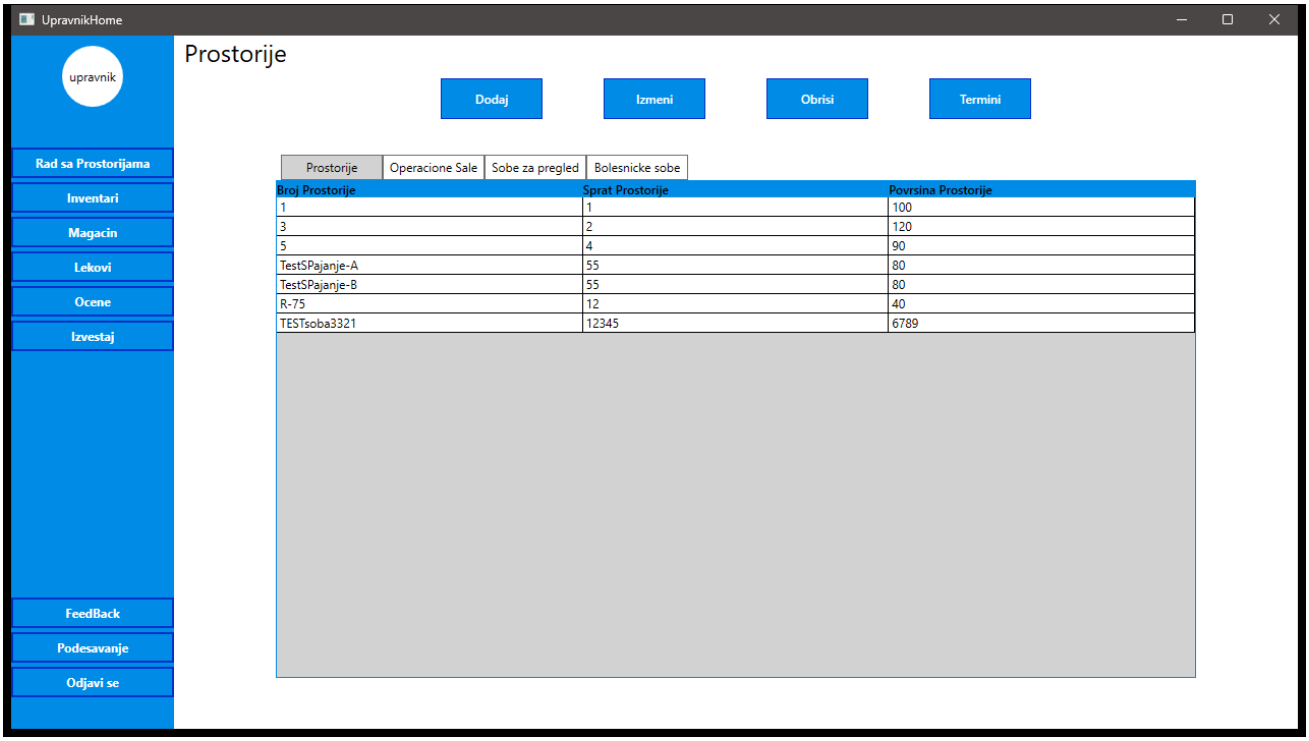


Figure 8. Manager rooms view (same Prostorije surface after sidebar click).

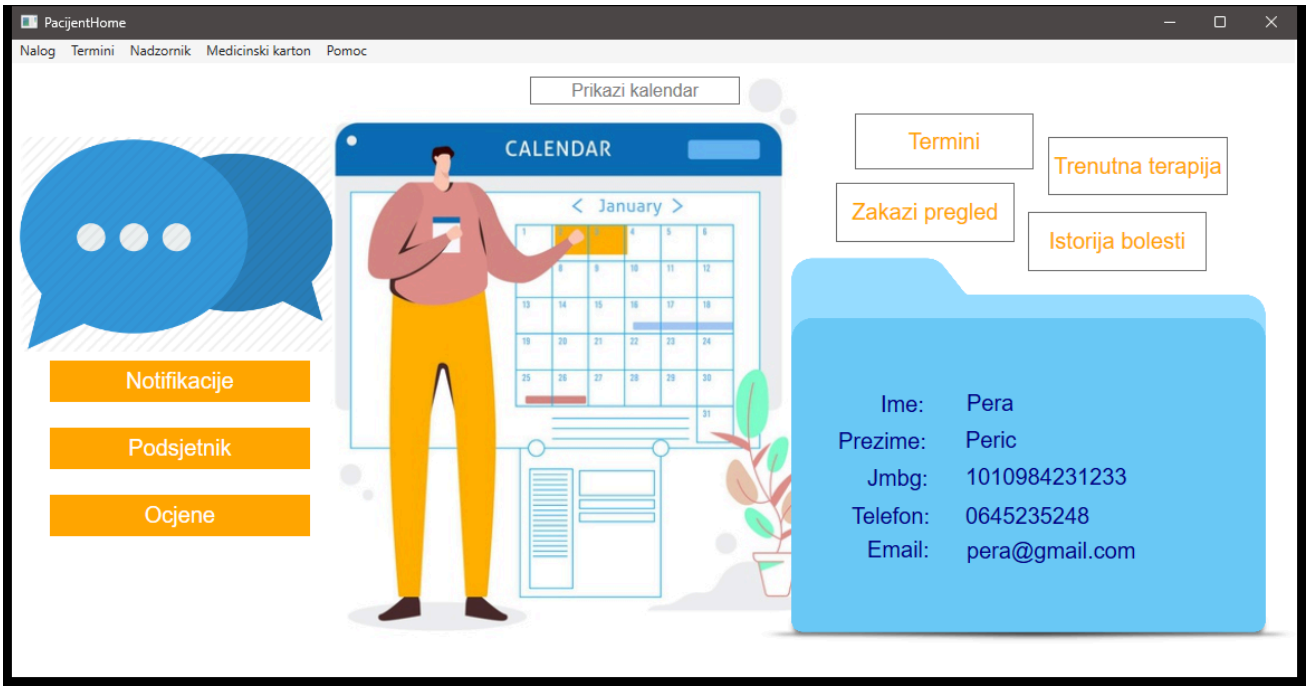


Figure 9. PacijentHome for Pera Peric — menus Nalog / Termini / Nadzornik / Medicinski karton / Pomoc; notifications, calendar graphic, Zakazi pregled, folder card with JMBG.

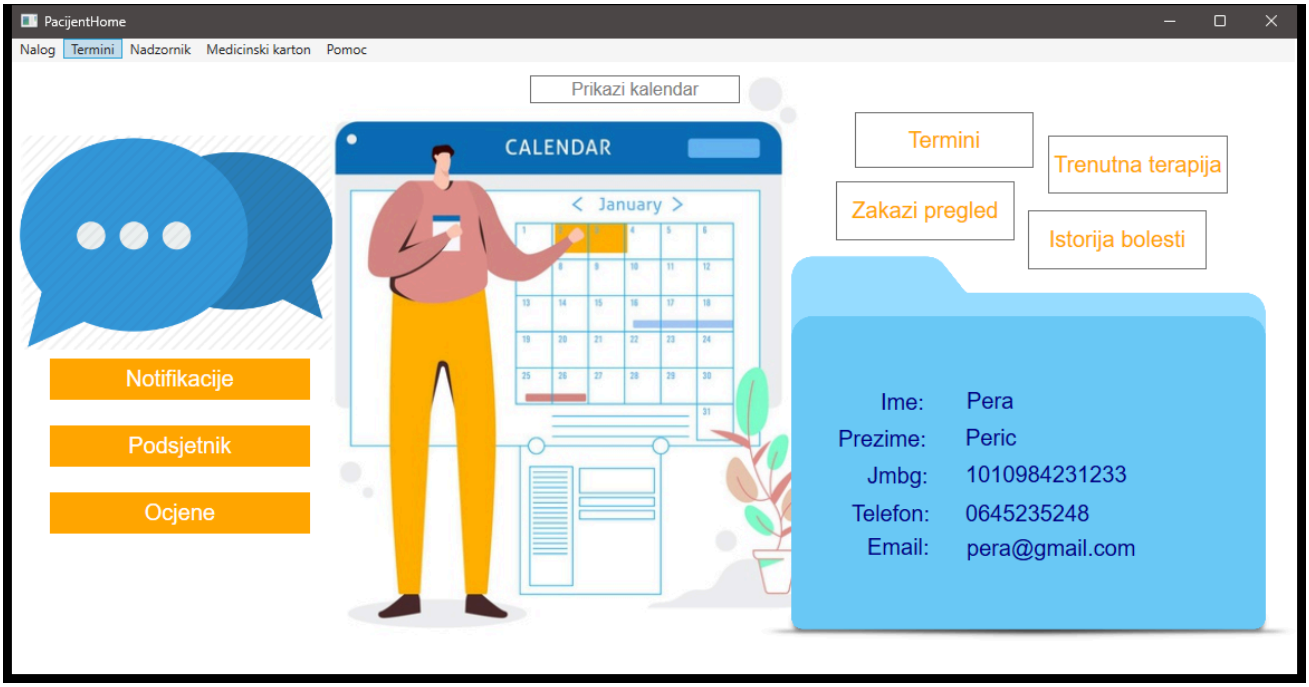


Figure 10. Patient shell after Termini navigation.

`MainWindow.Prijava_Click` tries secretary → manager → doctor → patient **in that order and does not return early**, so colliding usernames can open more than one home. Hidden debug buttons exist on the login XAML (`Visibility="Hidden"`) for some shells; `Lekar_Home_Click` is an empty stub.

Per-shell sizes as shipped: login ~450×600; doctor ~400×700; secretary ~1024×768; manager ~1280×720; patient ~1167×609.

6. Functional specification

Four roles share one store.

Upravnik (manager). Rooms by type (ordinary, OR, exam, ward): CRUD; schedule renovation, split, merge, equipment transfer (`TerminProstorije`). Per-room inventory (static / dynamic); reallocate between rooms; background thread applies transfers when appointments elapse. Warehouse (magacin). Drug catalogue with **doctor review** queue (`RevizijaLeka`). Ratings; PDF / chart doctor-work report; app feedback; settings (`podesavanja.json`). Live seed shows weak validation (absurd floor/area values left in `prostorije.json`).

Sekretar (secretary). First-run wizard and contextual HTML help (typos in wizard copy). Register patients (including guests) and doctors (MVVM on those forms). Browse/edit patients and kartons; doctors, work hours, vacation. Schedule exam/operation (multi-step wizard); emergency appointments; list/cancel/postpone. Announcements. Room occupancy. Rate the application; account page.

Lekar (doctor). Glavni meni → work calendar (day / week / month) over `ZdravstvenaUsluga`: record visit + anamnesis, update, delete. Patient karton: basics and allergens, therapy, inpatient referral, referral to another doctor, schedule a service, issue prescription. Medicines: approved list; revision queue; statistics/reports. Feedback. Some sidebar labels elsewhere in the tree are stubs or miswired.

Pacijent (patient). Book slots; view/cancel/reschedule; calendar (`Jarloo.Calendar`). Anti-spam: excess schedule/reschedule → `spamUser` and ~14-day suspension (`UserSuspension`). Notifications; medication reminders (`Podsjetnik`). Karton: therapy, history, adherence PDF. Prescriptions; hospital/doctor surveys; feedback; help overlays; keyboard shortcuts. Account change UI present.

Unique keys vary by repository (numeric `Id`, usernames for login). Drug codes use a `Sifra` field that is **not** a password.

7. Use cases

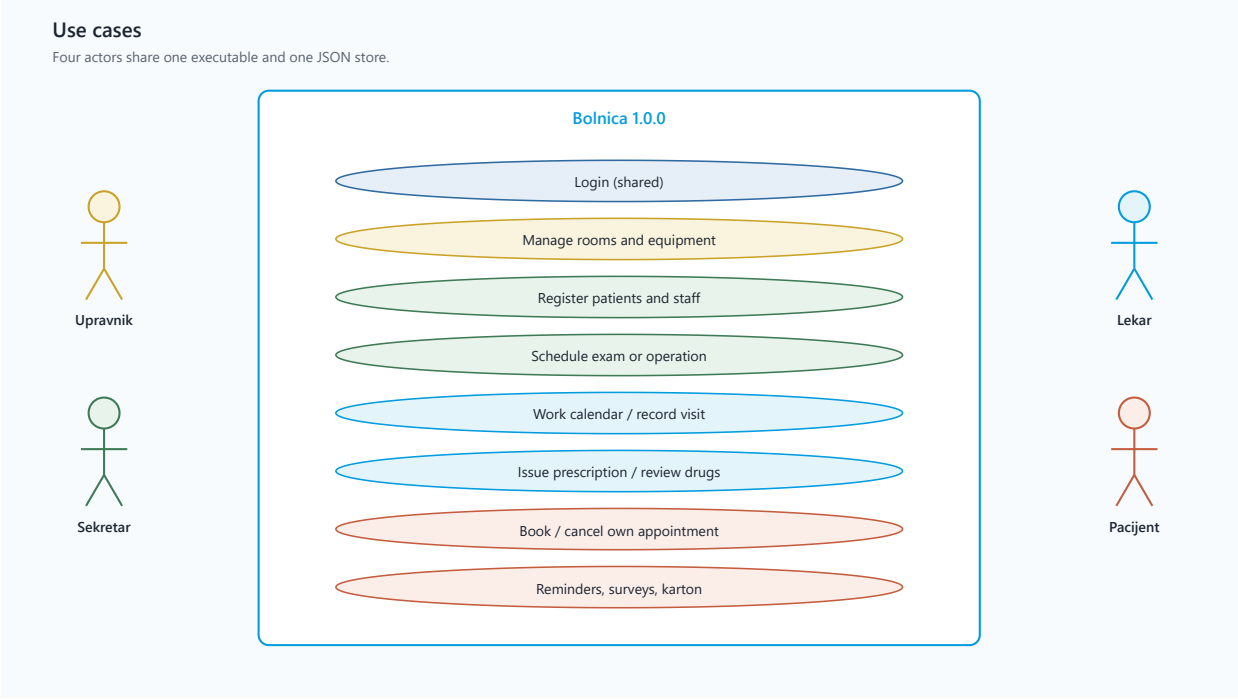


Figure 11. Four actors: Upravnik, Sekretar, Lekar, Pacijent. Login is shared.

Original PowerDesigner use-case drawings remain under `_legacy-1.0.0/Dijagrami/UseCase/`. This SVG is the dossier summary, not a conversion of those `.oom` files.

8. Domain model

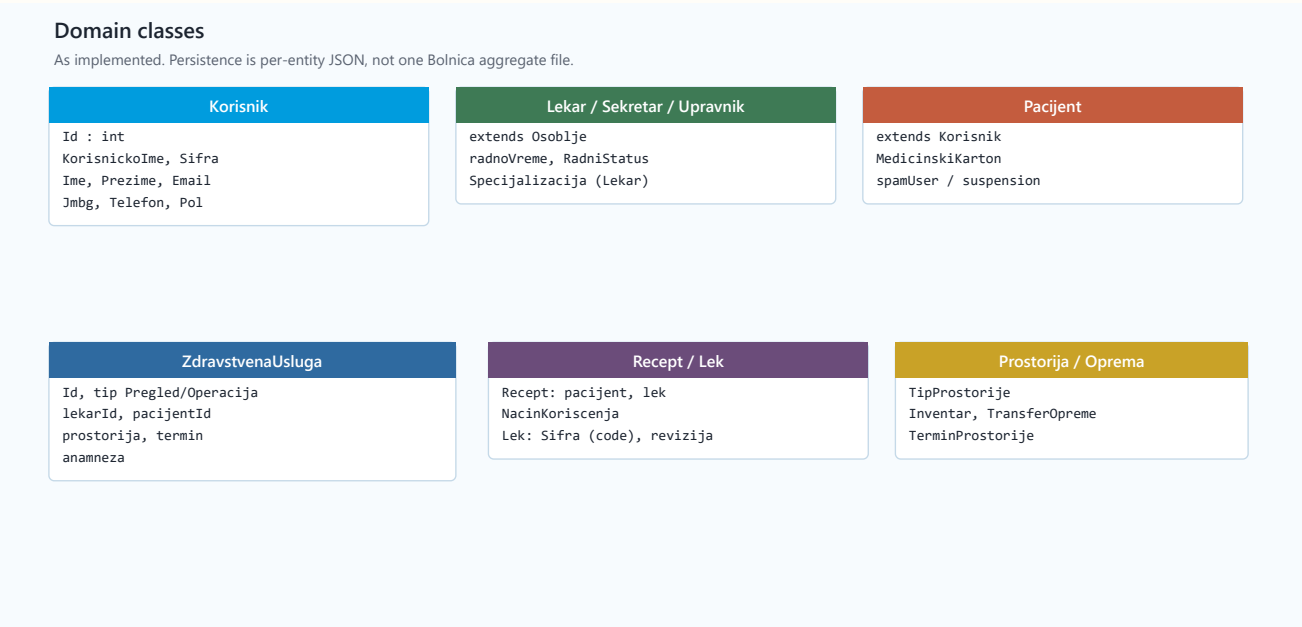


Figure 12. Core types as implemented.

Users: `Korisnik` → `Pacijent` | `Osoblje` → `Lekar`, `Sekretar`, `Upravnik` (plus unused `Nadzornik`).

Clinical: `MedicinskiKarton`, `ZdravstvenaUsluga`, `Termin`, `Terapija`, `Recept`, `Hospitalizacija`, `Lek`, `RevizijaLeka`.

Facility: `Prostorija`, `TerminProstorije`, `Inventar`, `Oprema`, `TransferOpreme`, `TransformacijaProstorijeParametri`.

Ops: `Obavestenje`, `Notifikacija`, `Podsjetnik`, `Feedback`, `Ocena`, `Aktivnost` (anti-spam), `RadnoVreme`, `GodisnjiOdmor`, `SlobodniDani`.

`Bolnica` exists as a model type; persistence is still per-entity JSON, not one aggregate file.

9. Architecture



Figure 13. View → Controller → Service → Repository → JSON.

Layer	Location	Notes
View	`view/lekar`, `sekretar`, `upravnik`, `pacijent`	~120 XAML surfaces; heavy code-behind
ViewModel	`viewmodel/`	Only three: patient/doctor registration, announcements
Controller	`Controller/` (`Kontroler`)	Role and domain façades
Service	`Service/`	Business rules
Repository	`Repository/`	Singletons `GetInstance`, load/save JSON
DTO	`DTO/`	Most flows
Patterns	Factory, Observer, Template Method	`BazaFactory` / `JSONBaza`; suspension & medication threads; report creators
Background	`Threads/`	User suspension, medication reminders, inventory reallocation

Not full MVVM. SQL factory branch is empty; only JSON is live. Freeze lives entirely under `_legacy-1.0.0/`.

10. Key sequences

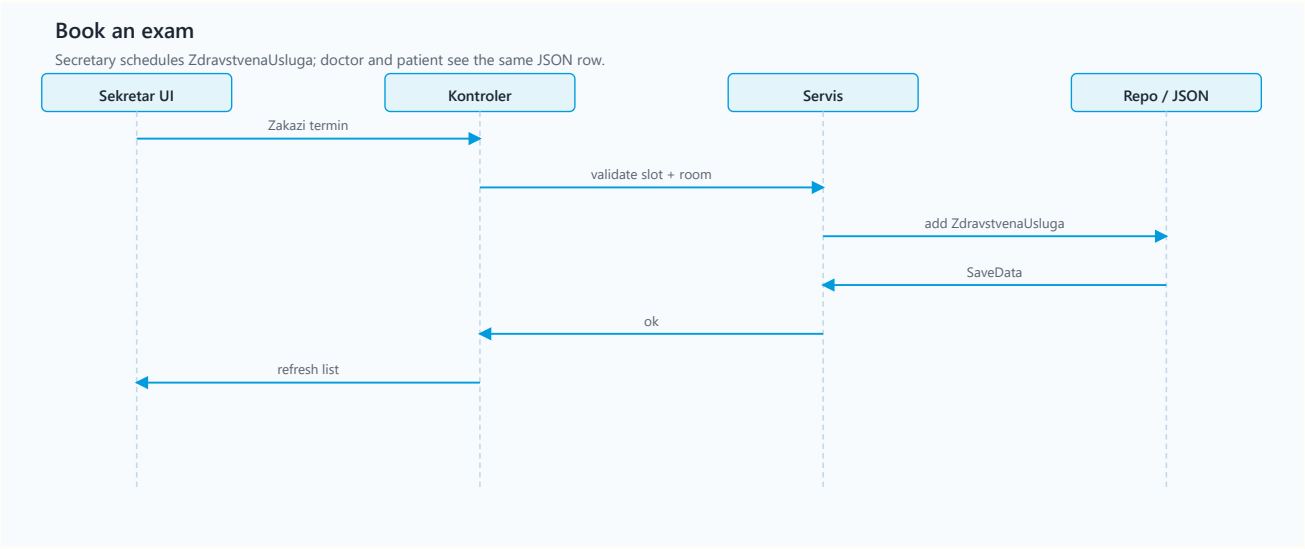


Figure 14. Secretary books an appointment → `zdravstveneUsluge.json` → doctor calendar / patient list.

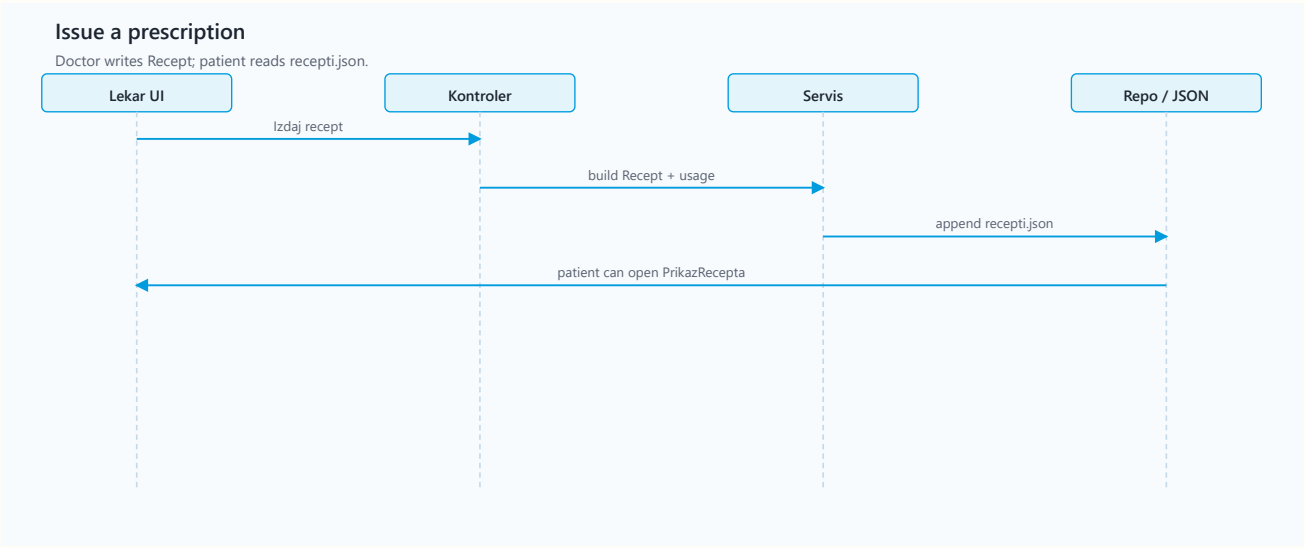


Figure 15. Doctor issues a prescription → `recepti.json` → patient view.

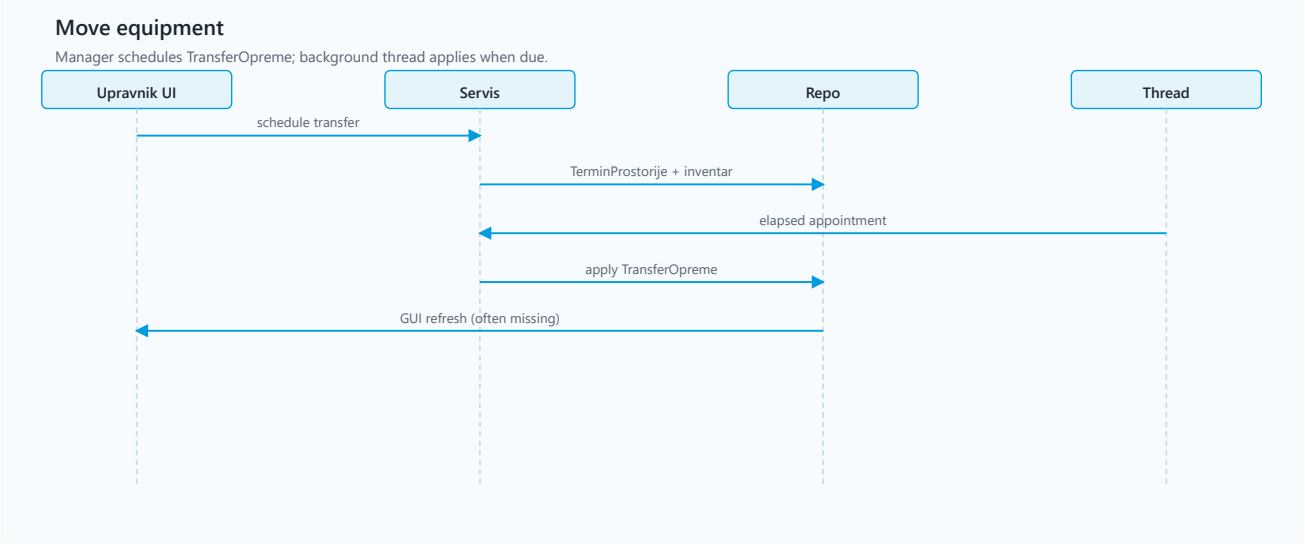


Figure 16. Manager schedules equipment transfer; thread applies when due. Peer review noted missing GUI refresh.

Course sequence drawings stay in `_legacy-1.0.0/Dijagrami/dijagram sekvence/` and `ZdravoBolnicaProj/`.

11. Persistence

Repositories keep lists in memory and call `SaveData()` with `System.Text.Json`. Paths are relative (`../../podaci/` from `bin\Debug`), resolving to `_legacy-1.0.0/BolnicaProjekat/Bolnica/podaci/`. No migrations, no file locking, no backup beyond overwriting the JSON file.

File	Approx. seed rows	Domain
`pacijenti.json`	7	Patients (+ embedded karton)
`lekari.json`	10	Doctors
`sekretari.json`	2	Secretaries
`upravnici.json`	1	Managers
`zdravstveneUsluge.json`	22	Appointments / services
`prostorije.json`	17	Rooms (includes test junk)
`terminProstorije.json`	2	Room appointments
`inventari.json`	18	Room inventories
`lekovi.json`	5	Medicines
`recepti.json`	9	Prescriptions
`obavestenja.json`	4	Announcements
`notifikacije.json`	19	Notifications
`podsjetnik.json`	29	Medication reminders

File	Approx. seed rows	Domain
`hospitalizacija.json`	3	Inpatient
`godisnjiodmor.json`	5	Vacation
`slobodnidani.json`	6	Days off
`feedback.json`	2	App feedback
`anketa.json`	5	Surveys
`podesavanja.json`	5	Settings

`data/sve.txt` is a small resource, not the primary store. Activity / anti-spam may create additional files at runtime.

12. Validation and security

Area	What is coded
Passwords	Clear text `Sifra` in JSON; string compare on login
Session	Static `App.ulogovaniKorisnik` / `App.IdUlogovanogKorisnika`
Authorisation	Role home opened after matching that role's repository; no process isolation
Login order	Secretary → manager → doctor → patient; **all four checks run**; multiple homes possible on collisions
Hashing / HTTPS / audit	None
Form validation	Incomplete; manager room seed proves absurd numbers are accepted
Copy quality	Secretary wizard: *aplikacijum*, *dogme*, *ocenivanj*

Demo credentials in the appendix are already public in the seed files.

13. Problems and technical debt

These belong to commit `5b69c9c`. They are here because the file is for a rewrite.

Area	What is wrong
Security	Plaintext passwords; static session globals; login can open more than one role home if usernames collide across JSON files
Persistence	Relative `../../podaci/` brittle; no migrations; no locking; overwrite-only saves; junk test rooms left in seed
Architecture	Incomplete MVVM (three ViewModels); empty SQL factory branch; Serbian/English identifier mix (`Kontroler` vs `Repository`)

Area	What is wrong
UI	Window title `MainWindow`; doctor empty-calendar “today” in 2026; secretary wizard typos and oversized overlay; manager table leaves empty lower pane; patient buttons loosely aligned
Doctor stubs	`Lekar_Home_Click` empty; some menu items elsewhere miswired
Domain / diagrams	Peer reviews (`Revizije/`): missing GUI refresh after equipment move; missing method names on sequences; anti-troll “suspension expired” flag not clearly in code; renovation activity I/O gaps
Concurrency	One process only; “four apps” marketing vs one EXE
Build in 2026	System 4.7.2 targeting pack often missing on Build Tools-only machines; NuGet `Microsoft.NETFramework.ReferenceAssemblies.net472` used as workaround (`run-1.0.0.bat`)
Gaps for a successor	No hashed auth, no real multi-user server, no SQL, few automated tests, no i18n

14. Implications for a rewrite

Keep the domain

- Four roles; appointments (exam / operation); prescriptions and drug review; rooms + equipment transfers; patient self-booking with anti-spam; surveys and reminders

Replace the machinery

- WPF 4.7.2 + JSON files with a current stack, hashed credentials, and a real schema
- Singleton repositories with migrations and backups
- Partial MVVM / code-behind with a consistent UI architecture and real window titles
- In-process “four personas” with either true multi-user services or an honest single-desk design

Do not copy

- Plaintext `Sifra` in seed JSON as a security model
- Login that continues checking after a match and can open multiple shells
- Empty SQL stubs that look like dual persistence
- Stub menu handlers and junk seed rooms
- Wizard copy typos as “features”

15. Build and 2026 check

Specified: Visual Studio 2019 era, .NET Framework 4.7.2, NuGet packages under `_legacy-1.0.0/BolnicaProjekat/Bolnica/packages/`.

Checked **7 September 2026** on Windows with **VS 2022 Build Tools** MSBuild. System targeting pack for 4.7.2 was absent; build succeeded using NuGet reference assemblies (`Microsoft.NETFramework.ReferenceAssemblies.net472` / `net40`) and `FrameworkPathOverride` (also wired into `run-1.0.0.bat`).

```
cd _legacy-1.0.0\BolnicaProjekat\Bolnica
nuget restore Bolnica.sln
msbuild ..\Jarloo.Calendar\MyCalendar.Calendar.csproj /p:Configuration=Debug
/p:FrameworkPathOverride=...\net40
msbuild Bolnica.csproj /p:Configuration=Debug /p:FrameworkPathOverride=...\net472
cd bin\Debug
Bolnica.exe
```

Repo root: `run-1.0.0.bat`. Login with Appendix A accounts produced the screenshots in section 5. Doctor Današnji calendar was empty (seed appointments are not relative to 2026 “today”).

Appendix A. Seed users (plaintext passwords as shipped)

Username	Password	Role	Notes
`rastislavkukucka`	`sifra`	Upravnik	Sole manager seed
`mihajloostojic`	`sifra`	Sekretar	
`mihajlo`	`sifra`	Sekretar	Second secretary
`LasloUri`	`laci`	Lekar	Course doctor account
`JovanaJovanovic`	`joka`	Lekar	
`nikolanikolic`	`sifra`	Lekar	Username also appears as a patient with a different password
`peraperic`	`pera`	Pacijent	Captured in Figure 9
`nikolanikolic`	`nikola`	Pacijent	
`simasima`	`sifra`	Pacijent	
`markomarkovic`	`asdasd`	Pacijent	
`jokajovanovic`	`sifra`	Pacijent	

Additional doctor rows use passwords like `sifra` / `sifra2` / `sifra3` as shipped.

Appendix B. JSON file map

See section 11. Working directory for runtime should keep `../../podaci/` resolving to `_legacy-1.0.0/BolnicaProjekat/Bolnica/podaci/`.

Appendix C. Git freeze

Item	Value
Tip	`5b69c9c` · 2021-06-21 · *test - git nalog*
Root	`8fe447c` · 2021-03-16 · Initial commit
Commits	305

Item	Value
Origin now	https://github.com/laslo-uri/HospitalManagementSuite.git
Upstream	https://github.com/kukuckarastislav/SIMS_IS_Bolnica
Layout 2026	Freeze under `_legacy-1.0.0/`; docs at repo root

Appendix D. Launch this snapshot

1. Install Visual Studio or Build Tools with .NET desktop (or use NuGet reference assemblies as in `run-1.0.0.bat`).
2. From repo root: `run-1.0.0.bat`, or restore + build under `_legacy-1.0.0/BolnicaProjekat/Bolnica`.
3. Run `Bolnica.exe` from `bin\Debug`.
4. Log in with a seed account from Appendix A.
5. Optional: open `_legacy-1.0.0/Dijagrami/` in PowerDesigner; read `_legacy-1.0.0/Revizije/` for KT4 peer notes.

Appendix E. Course drawings and reviews

Path	Role
`_legacy-1.0.0/Dijagrami/UseCase/`	Per-role and overall use cases (`.oom`)
`_legacy-1.0.0/Dijagrami/ClassDiagram/`	Domain class diagram
`_legacy-1.0.0/Dijagrami/dijagram aktivnosti/`	Prescription, patient, emergency scheduling
`_legacy-1.0.0/Dijagrami/dijagram sekvence/`	Referral, patient, secretary sequences
`_legacy-1.0.0/Dijagrami/ZdravoBolnicaProj/`	Manager sequences / activities
`_legacy-1.0.0/Revizije/*.txt`	Peer reviews (Laslo reviewed manager equipment sequence; others reviewed prescription, scheduling, anti-troll)

Compiled 7 September 2026 from commit `5b69c9c` (21 June 2021). Dossier 1.0.0.